Balanced–MixUp On Manifold For Highly Imbalanced Breast Cancer Classification

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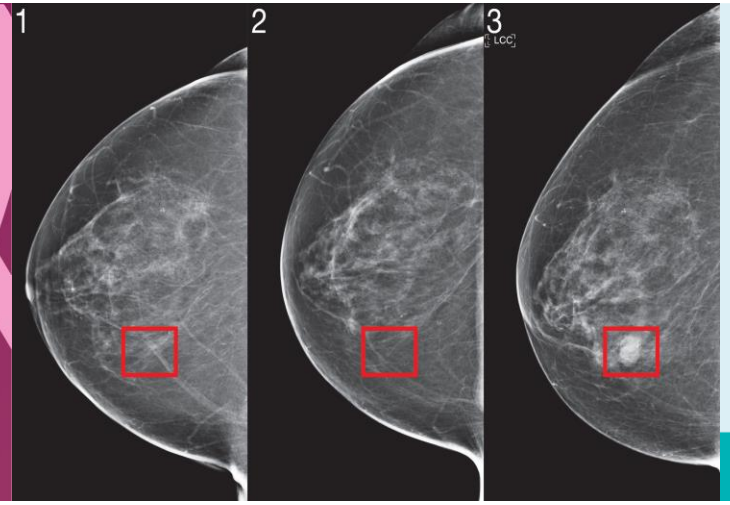
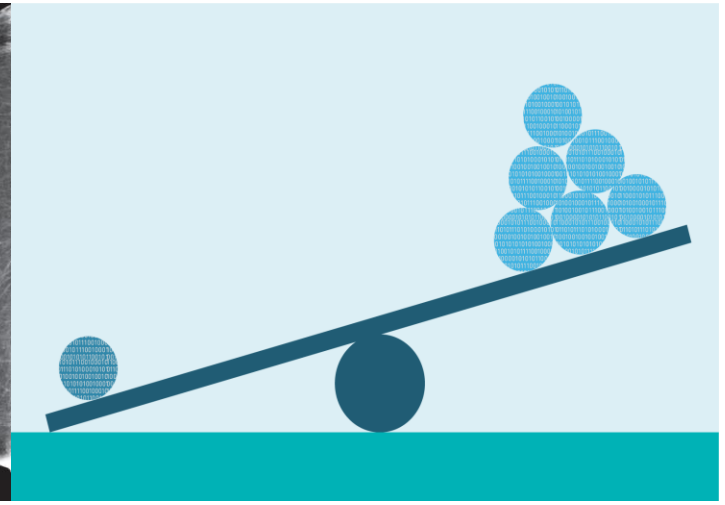
Background

- **Breast cancer** is one of the most frequently diagnosed cancers in women globally worldwide.
- **Deep learning** has become a prominent focus in the study and application of mammogram analysis for early breast cancer detection.
- **Challenges:** Normal images significantly outnumber abnormal ones, leading to model bias toward the majority class in training diagnostics models.





Motivation

- **Balanced-MixUp (Galdran et al., 2021**) is new imbalanced-robust training method that mixes up imbalanced (instance-based) and balanced (class-based) sampling of the data.
- **Manifold Mixup (Verma et al., 2019**) is a simple regularizer that encourages neural networks to make less confident predictions on interpolations of hidden representations, resulting in smoother decision boundaries and more compressed class representations.

Motivated by this (Balanced-MixUp on Manifold)

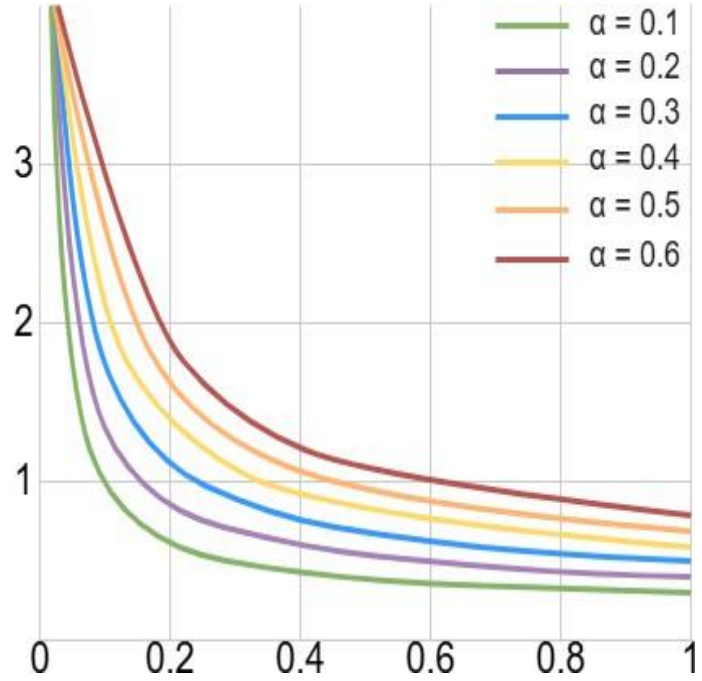
1. We propose a novel training strategy that combines Balanced-MixUp on Manifold to effectively address the challenges of highly imbalanced data in breast cancer classification.
2. We present Balanced-MixUp on Manifold loss that integrates two powerful regularization techniques to address both data imbalance and feature-level generalization challenges.

Our Approach: Balance MixUp on Manifold

Balance MixUp: Instance-based sampling (imbalanced) and Class-based sampling (balanced)

1. Instance-based sampling : A data sampling method where the probability of selecting a sample from a class is proportional to the frequency of that class in the training dataset (q=1). In other words, samples are drawn randomly according to the original class distribution.
2. Class-based sampling : A data sampling method where each class is selected with equal probability, i.e., uniform sampling (p_j = 1/K, where K is the number of classes), resulting in oversampling of minority classes to balance the class distribution during training (q=0).

Notation : $\hat{x} = \lambda x_I + (1 - \lambda) x_C,\ \hat{y} = \lambda y_I + (1 - \lambda) y_C$



Beta(α, 1) distribution for varying α

Our Approach: Balance MixUp on Manifold

Manifold Mixup: performs linear interpolation on hidden representations at various layers within the neural network during training, not just in the input space as in Input Mixup, and this interpolation is also applied to the labels with corresponding soft labels.

1. Layer Selection: A layer *k* was randomly selected from a predefined set of eligible layers *S* within the network, typically from mid to deep layers where feature abstraction is high.
2. Mini-Batch Sampling: Two separate mini-batches (*x, y*) and (*x′, y′*) were processed through the network up to layer *k*. Here, $x, x' \in R^{H \times W \times C}$ represent the image inputs, and $y, y' \in R^{K}$ are the corresponding one-hot encoded labels.
3. Latent Feature Interpolation: The intermediate feature representations $g_k(x)$ and $g_k(x')$ were linearly combined:

$$\tilde{g}_k \coloneqq \lambda \cdot g_k(x) + (1 - \lambda) \cdot g_k(x'), \qquad \tilde{y} \coloneqq \lambda \cdot y + (1 - \lambda) \cdot y',$$

where λ Beta(α, α) controls the interpolation strength.

Our Approach: Balance MixUp on Manifold

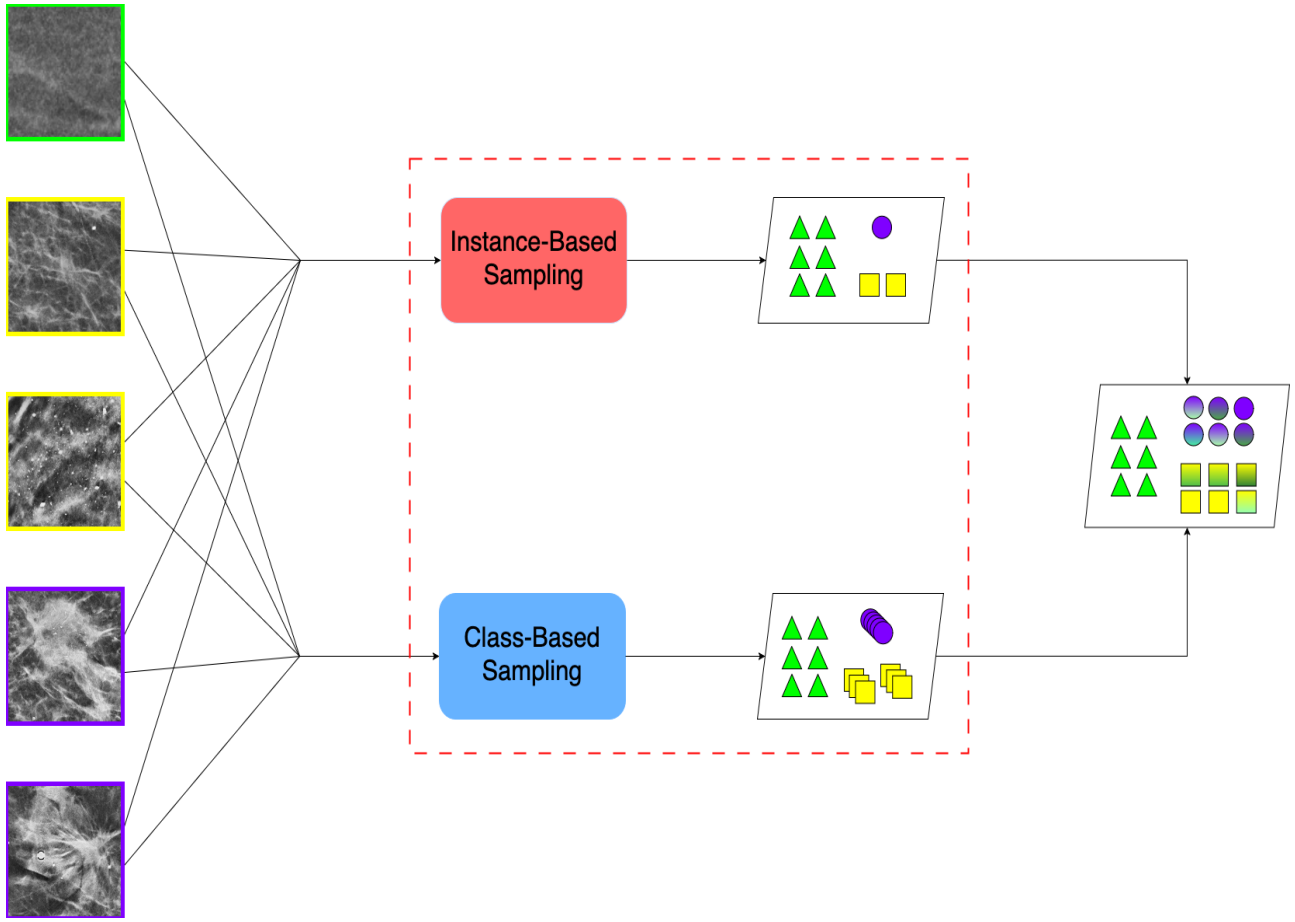
4. Forward Pass & Training: The mixed representation $\tilde{g}_k$ was passed through the remaining layers f_k to generate predictions. The model was trained using the mixed labels $\tilde{y}$ with cross-entropy loss:

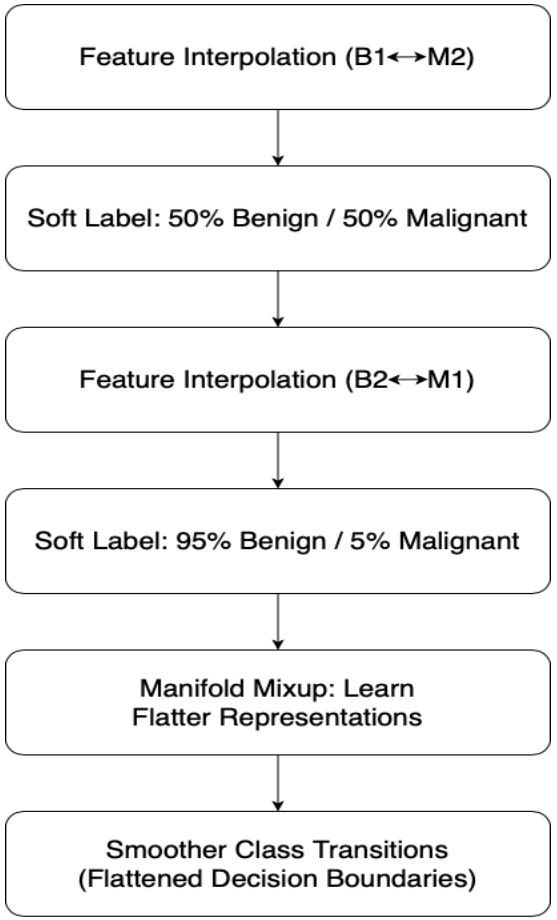
$$\mathcal{L}(\mathrm{f}) = \mathrm{E}_{(\mathrm{x},\mathrm{y}),(\mathrm{x}',\mathrm{y}')}\, E_\lambda E_k \ell(f_k)(\tilde{g}_k), \tilde{y}),$$

   where ℓ denotes the loss function (e.g., cross-entropy).

5. Gradient Backpropagation: Gradients were propagated through entire network. allowing the model to benefit from the regularized latent representations.

Our Approach: Balance MixUp on Manifold





Our Approach: Balance MixUp on Manifold

Loss Function

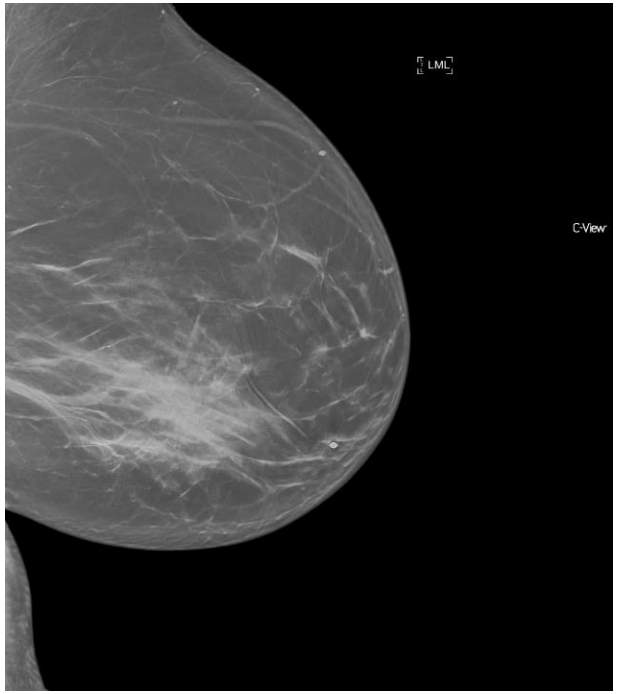
Balance MixUp on Manifold, novel loss computation strategy that improves robustness by interpolating not only inputs but also intermediate feature representations, resulting in improved discrimination, especially in highly imbalanced datasets like breast cancer classification.

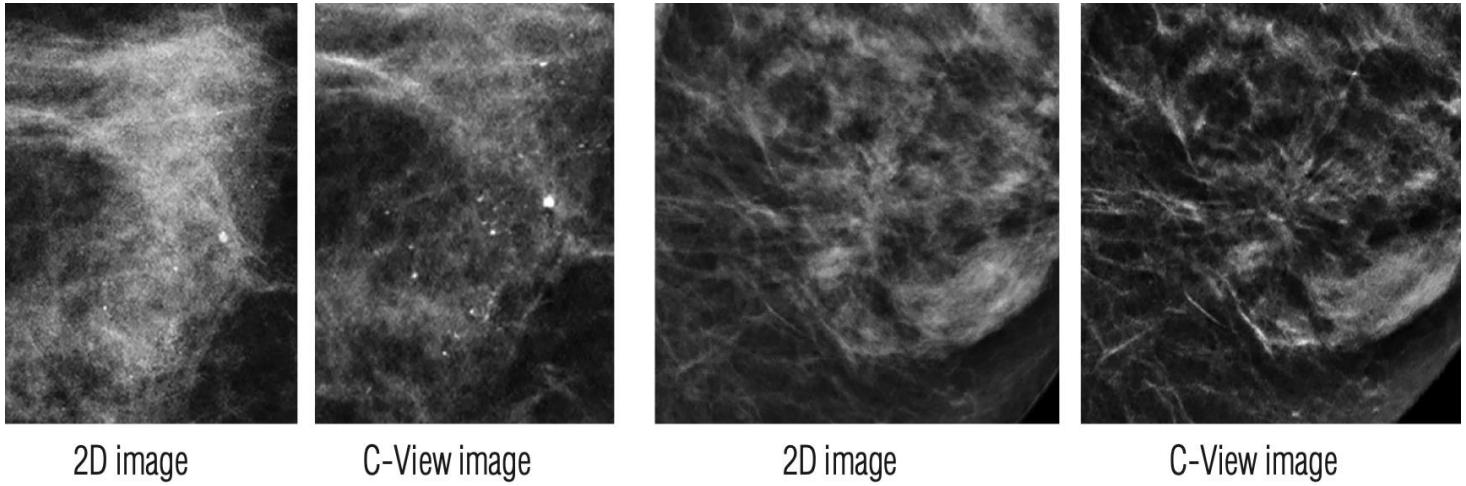
Notation

$$\mathcal{L}_{\mathrm{BM+MM}} = \lambda \cdot \mathcal{L}_{\mathrm{CE}}\left(f\left(x_a^h\right), y_a\right) + (1-\lambda) \cdot \mathcal{L}_{\mathrm{CE}}\left(f\left(x_b^h\right), y_b\right)$$

Dataset

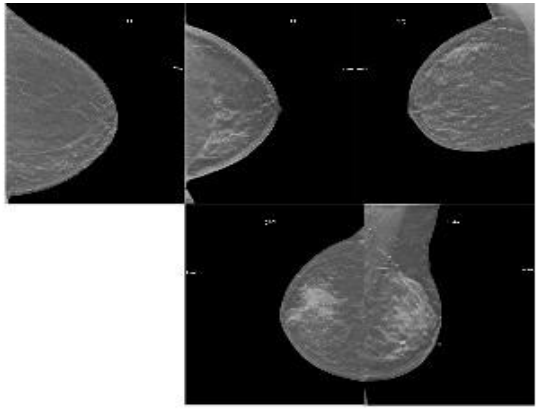
- We use C-view Mammogram of the EMory BrEast Imaging Dataset (EMBED) (https://github.com/Emory-HITI/EMBED_Open_Data)

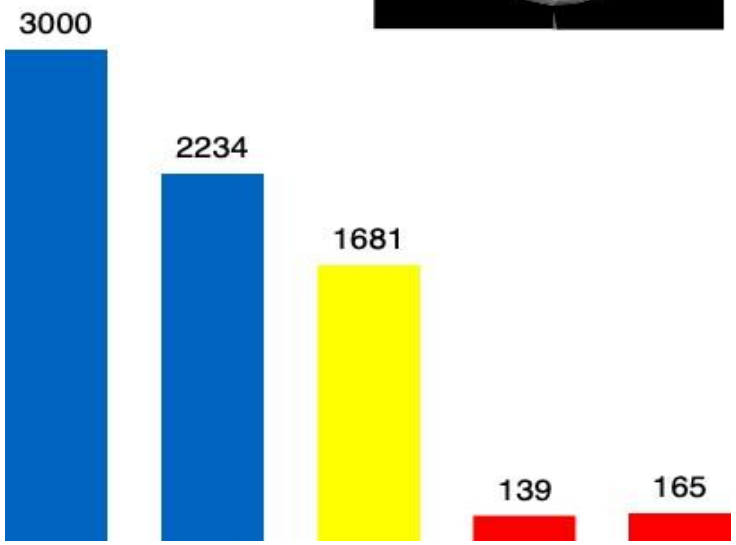




Dataset

- The mammogram dataset includes BI-RADS categories:
  1. BI-RADS 1: Negative = 3000
  2. BI-RADS 2: Benign = 2234
  3. BI-RADS 3: Suspicious = 1681
  4. BI-RADS 5: Highly Suggestive of Malignancy = 139
  5. BI-RADS 6: Known Biopsy-Proven Malignancy = 165
- BI-RADS 1,2 are categorized into Benign (non-cancer) and BI- RADS 3,5,6 are categorized into Malignant (cancer)



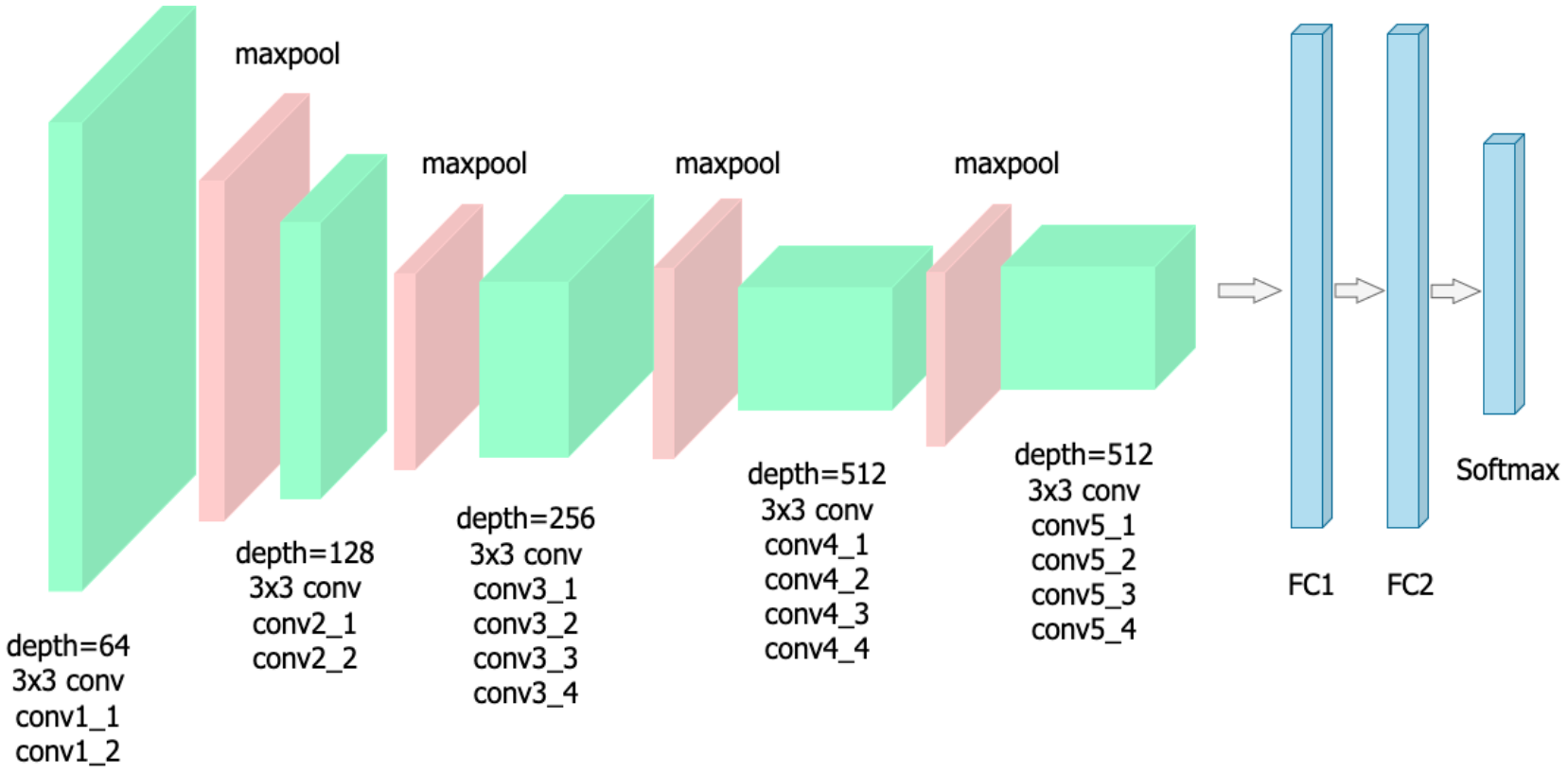


Dataset

BI-RADS Categories	Training (64%)	Validation (16%)	Testing (20%)
BI-RADS 1: Negative	1920	480	600
BI-RADS 2: Benign	1430	357	447
BI-RADS 3: Suspicious	1075	268	338
BI-RADS 5: Highly Suggestive of Malignancy	94	24	21
BI-RADS 6: Known Biopsy-Proven Malignancy	105	27	33
Total	4624	1156	1439

Model

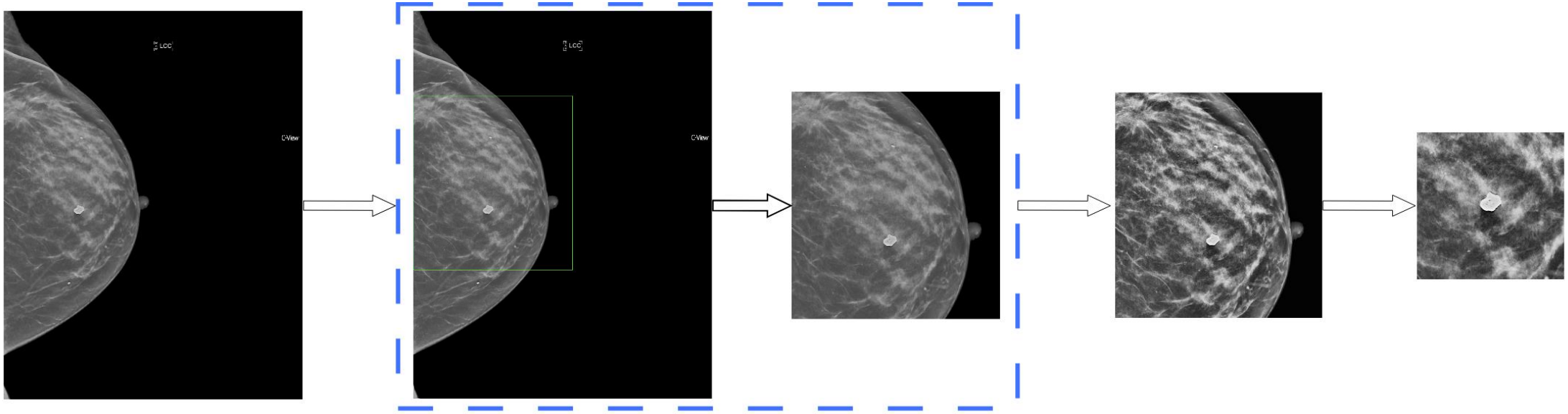
VGG-19


maxpool
maxpool
maxpool
maxpool
depth=64
3x3 conv
conv1_1
conv1_2
depth=128
3x3 conv
conv2_1
conv2_2
depth=256
3x3 conv
conv3_1
conv3_2
conv3_3
conv3_4
depth=512
3x3 conv
conv4_1
conv4_2
conv4_3
conv4_4
depth=512
3x3 conv
conv5_1
conv5_2
conv5_3
conv5_4
FC1
FC2
Softmax

Training Parameter

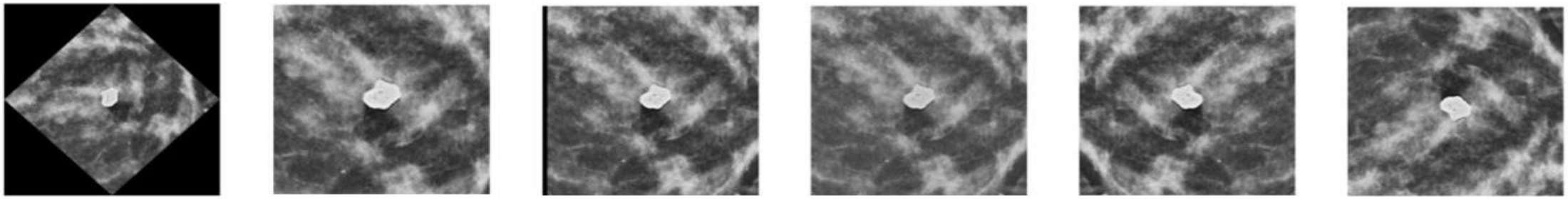
Framework	Loss Function	Optimizer	Batch Size	Learning Rate
Pytorch	Cross Entropy	SGD (Stochastic Gradient Descent)	8	0.01

Experiments (1/3)



(a) Converting format (b) Denoising & Cropping (c) CLAHE (d) Resizing

Experiments (2/3)



Rotate 45° | Scale1.2x | Translate 5% | Color Jitter | Horizontal Flip | Vertical Flip

Experiments (3/3)

	VGG-19		
	MCC	**B-ACC**	**Macro-F1**
Class-sampling	58.45	50.48	50.71
Instance-sampling	**61.39**	48.23	48.39
Sqrt-sampling	60.31	48.97	49.46
Focal Loss	59.93	45.74	43.78
CB Loss	58.74	45.68	43.51
Bal-Mxp: α = 0.1	60.91	47.32	46.58
Bal-Mxp: α = 0.2	59.31	50.16	50.87
Bal-Mxp: α = 0.3	58.50	43.90	41.40
Bal-Mxp: α = 0.4	58.06	47.38	47.09
Bal-Mxp: α = 0.5	60.23	**50.93**	**52.70**
Bal-Mxp: α = 0.6	58.73	46.74	46.59
Proposed : Balanced-MixUp on Manifold			
Bal-Mxp: α = 0.1	58.50	**50.84**	**49.91**
Bal-Mxp: α = 0.2	58.50	**51.56**	**52.94**
Bal-Mxp: α = 0.3	57.03	**55.53**	**54.44**
Bal-Mxp: α = 0.4	56.25	**52.64**	**53.32**
Bal-Mxp: α = 0.5	56.07	47.91	48.09
Bal-Mxp: α = 0.6	**58.91**	**51.33**	**51.66**

Table 1: Stratified 5-fold Cross-validation (Median) Of VGG-19 Results For Breast Cancer Classification On The EMBED. Best Results Are Marked Bold. Best Proposed Method Are Marked In Blue With Red Underline.

Conclusions

Summary

Our proposed method Balanced-MixUp on Manifold method consistently improved class imbalance handling & significant improvements in B-ACC and Macro-F1.

Limitation & Future Work

Interpolation strength α was manually tuned (0.1–0.6). Manual tuning is time-consuming and may not generalize well across models or datasets. Future work could explore adaptive mechanisms to dynamically adjust α during training.